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File Permissions and Special Permissions

Fundamentals of Information Technology | Session 8 | 4–5 November 2026

Learning Objectives

By the end of this session, you will be able to:

1. Read Linux permission strings such as -rwxr-xr--
2. Change permissions and ownership with chmod, chown, and umask
3. Explain and apply SUID, SGID, and the sticky bit
4. Apply least privilege and diagnose “Permission denied” errors

Today's Agenda

1. Users, groups, and others
2. Reading permission strings
3. Octal mode, chmod, chown, umask
4. Special permissions: SUID, SGID, sticky bit
5. Hands-on lab: a shared folder
6. Security and troubleshooting

Every file has an owner, a group, and rules for others

u **User (u)**

The owner of the file, usually the person who created it.

g **Group (g)**

Users who share a group, such as a project team.

o **Others (o)**

Everyone else who has an account on the system.

Users, Groups, Others

Permissions are set separately for each of the three.

Reading Permission Strings

`-`	`rwx`	`r-x`	`r--`
File type	**User (owner)**	**Group**	**Others**
- regular file d directory l symbolic link	Permissions for the owner of the file, here alice.	Permissions for members of the file's group, here staff.	Permissions for everyone else on the system.

```
-rwxr-xr-- 1 alice staff 4096 Nov 4 deploy.sh
```

What r, w, and x Mean

Permission	On a file	On a directory
r (read)	View the contents	List the names inside (ls)
w (write)	Modify or overwrite the contents	Create, delete, and rename entries (also needs x)
x (execute)	Run it as a program or script	Enter it (cd) and reach items inside

Deleting a file depends on the write permission of its directory, not of the file itself.

Octal Mode: r = 4, w = 2, x = 1

r	w	x
Read = 4	Write = 2	Execute = 1

Example: rwxr-xr--

rwx	r-x	r--
4 + 2 + 1 = 7	4 + 0 + 1 = 5	4 + 0 + 0 = 4

User, group, others = **754** → **chmod 754 deploy.sh**

Common Permission Modes

rwxr-xr-x

755

Programs and shared directories: owner full access, others read and run.

rw-r--r--

644

Regular files such as documents and web pages.

rwx------

700

Private directories that only the owner can enter.

rw-------

600

Private files such as SSH keys and credentials.

rw-r-----

640

Owner edits, group reads, others have no access.

rwxrwxrwx

777

Everyone can do everything. Avoid it: it is a security risk.

chmod, chown, and umask

chmod: change permissions

```
chmod 644 notes.txt
chmod u+x deploy.sh
chmod g-w,o-rwx secret.txt
chmod -R 755 website/
```

u user, g group, o others, a all. + adds, - removes, = sets exactly. -R applies to everything inside.

Ownership and defaults

```
sudo chown alice file.txt
sudo chown alice:staff file.txt
sudo chgrp staff file.txt
umask 022
```

Only root can give a file away with chown. umask 022 makes new files 644 and new directories 755.

Shared Folder Lab

Needs a Linux terminal: WSL, a virtual machine, or a lab PC. Work inside the Linux home folder, not on a Windows drive.

Expected result

```
$ ls -ld shared
drwxrwsr-t 2 alice staff 4096 Nov 4 shared
```

STEP 01

Create

Run mkdir shared, then chmod 775 shared (rwx for owner and group, r-x for others).

STEP 02

Add the bits

Run chmod g+s shared for SGID, then chmod +t shared for the sticky bit.

STEP 03

Verify

Run ls -ld shared and look for the s and the t in the result.

Permission Best Practices

Least privilege

Give only the access that is needed. Prefer 644 and 755, and avoid 777.

Audit SUID and SGID

Run find / -perm -4000 regularly and investigate anything unexpected.

Careful with SUID

SUID on a shell, editor, or interpreter can hand out root access.

Protect secrets

Keep SSH private keys and credential files at 600.

Troubleshooting Permission Problems

Symptom	Likely cause	What to do
“Permission denied” running a script	Missing x bit on the file	chmod u+x script.sh
Cannot list a directory	Missing r on the directory	Check ls -ld dir and add r
cd into a directory fails	Missing x on the directory	Add x, for example chmod 755 dir
Cannot delete a file you own	No w on the parent directory, or sticky bit	Check the directory with ls -ld
chmod: Operation not permitted	You are not the owner	Use sudo or ask the owner
Permissions will not change on a USB drive	FAT32, exFAT, or NTFS has no Linux permissions	Use ext4 or set mount options

Summary

1. Every file has an owner, a group, and rules for others. Read them with ls -l.
2. r, w, and x mean different things on files and directories. In octal: 4, 2, 1.
3. chmod changes permissions, chown and chgrp change ownership, umask sets defaults.
4. SUID, SGID, and the sticky bit are special permissions shown as s and t.
5. Use least privilege and audit SUID files regularly.

Exercises and What's Next

Case study

Alice creates report.sh but gets “Permission denied” when she runs ./report.sh. Its mode is 644. How do you fix it, and what should the mode be?

Quick discussion

- What does the s in -rwsr-xr-x mean?
- Why can't users delete each other's files in /tmp?
- Why is chmod 777 risky?

Session 9 | 11–12 Nov 2026

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Networking Concepts

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